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1. Introduction

This design specification describes the functionality, API, and the design decisions for CddTeleBackupMngr Component that will be used in CSAR.

This design for this document is as per the requirements mentioned in the Integrity document id < 9167284 >: name < CddTeleBackupMngr>.

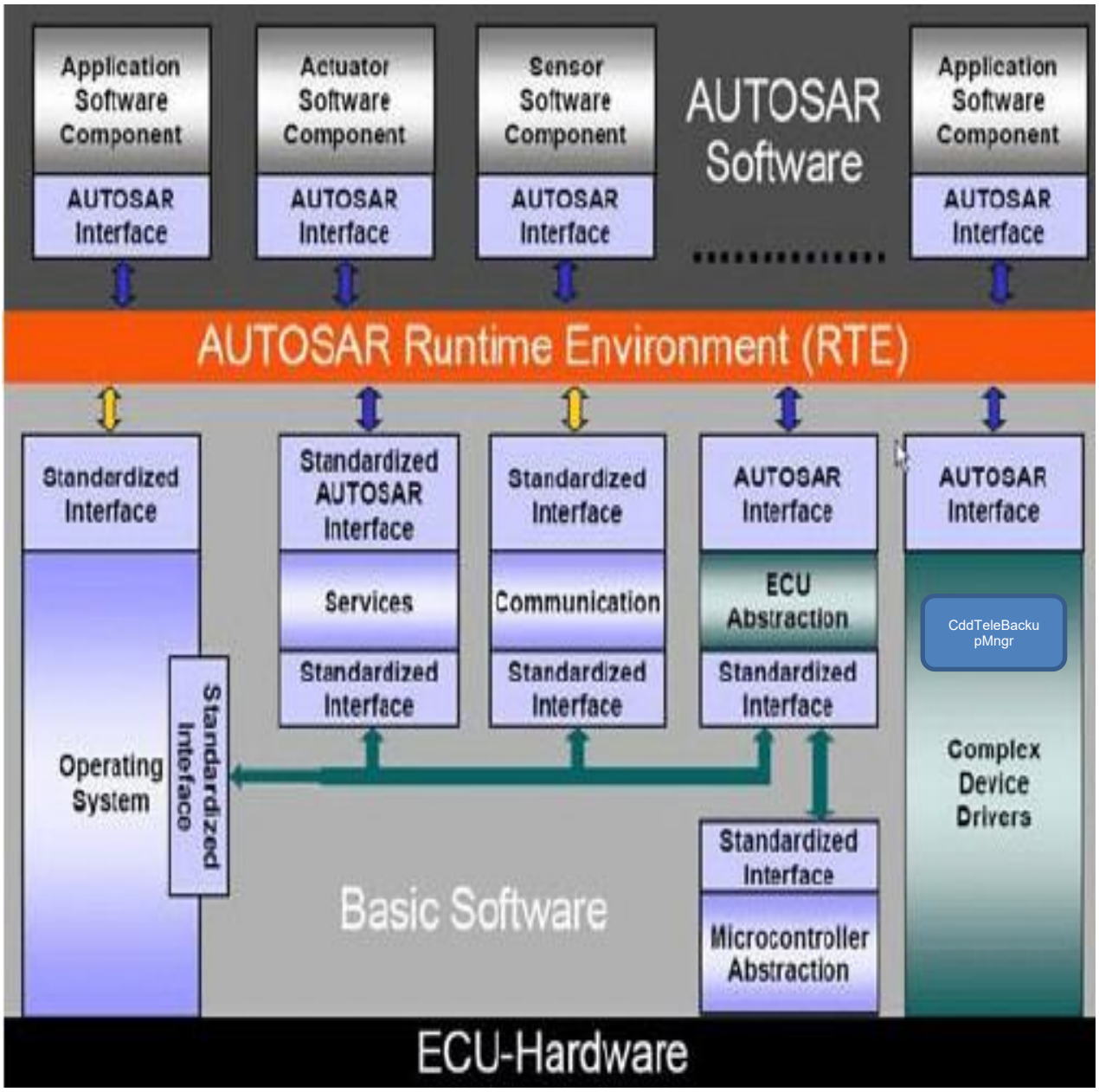


Figure 1: CDDTeleBackupMngr CSAR Layer Model

2. Acronyms and Abbreviations

The glossary below includes acronyms and abbreviations relevant to < CddTeleBackupMngr > that are used in this document.

Abbreviation / Acronym:	Description:
BSW	Basic Software
SWC	Software Component
SWS	Software Specification
SDD	Software Detailed Design Document
JES	Journaling Entry System
CDD	Complex Device Driver
RTE	Run Time Environment
INIT	Initialization
PP	Provider Port
API	Application Program Interface
DET	Default Error Tracer
MCAL	Microcontroller Abstraction Layer
SAC	Sensor Actuator Component
ID	Identifier
CSAR	Cummins AUTOSAR

3. Bibliography

All documents of the referenced Specification of CSAR CSU Feature are also valid for this document.

3.1 Input documents & related standards and norms

[1] Layered Software Architecture AUTOSAR_EXP_LayeredSoftwareArchitecture

[2] Glossary AUTOSAR_TR_Glossary

[3] General Specification of Basic Software Modules
AUTOSAR_SWS_BSW_General

[4] AURIXTC3XX_um_part2_V1.0.0

[5] MC-ISAR_TC3xx_UM_Spi

[6] Data sheet of Cypress S25FL512S and
Micron 25QL512

3.2 Related requirement specification

A requirement document for CddTeleBackupMngr is stored in Integrity tool under the id < 9167284 >.

This design specification satisfied the requirements listed in the above requirement document for this SWC.

4. Functional specification

4.1 General Functionality

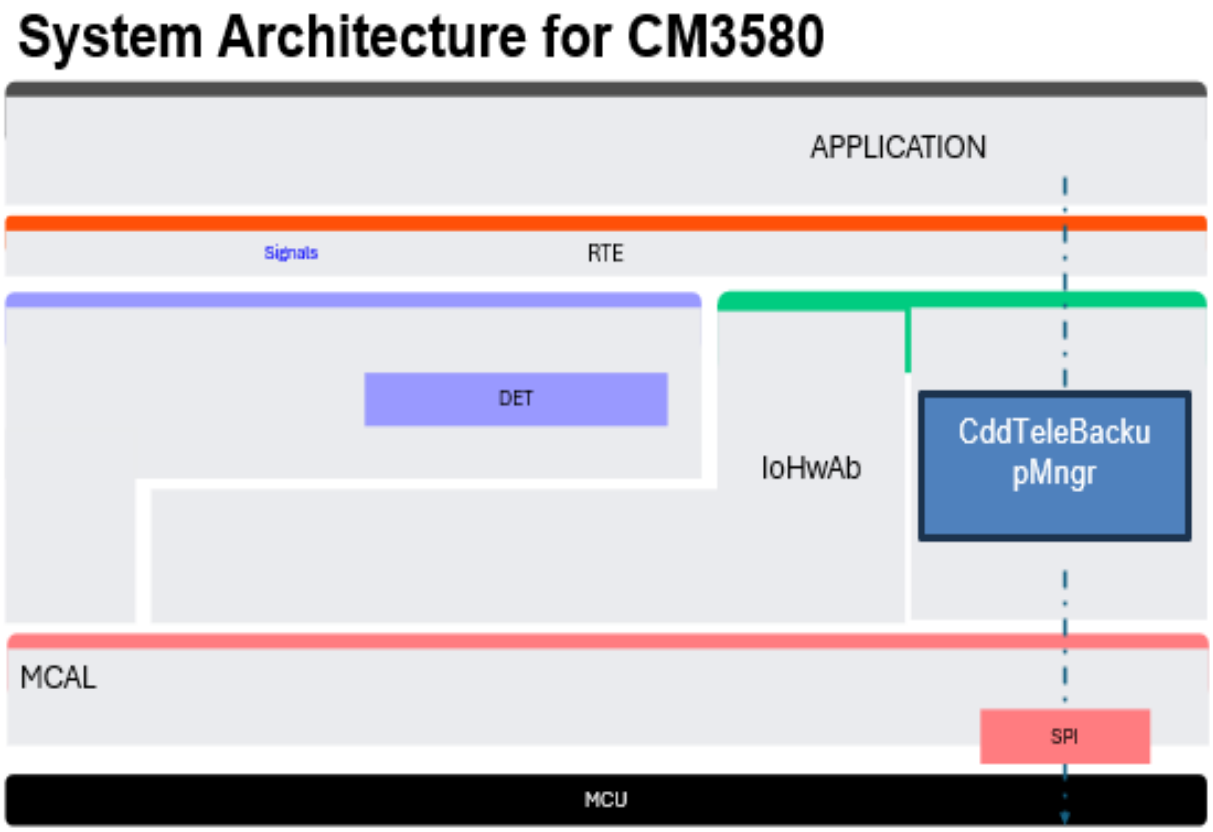
The module is a Complex Device Driver (CDD) designed within the CSAR (Cummins AUTOSAR) architecture to manage secure backup operations of program flash data into Serial Flash (SF). The module is responsible for handling backup requests, ensuring data integrity, confidentiality, and reliability during the complete backup CddTeleBackupMngr lifecycle.

The backup mechanism is implemented as a state-driven process, where each state performs a well-defined operation. The module supports encryption, authentication (CMAC), verification, and controlled retries in case of failures. Upon successful completion, backup status information is persistently stored in NVM.

Key Responsibilities

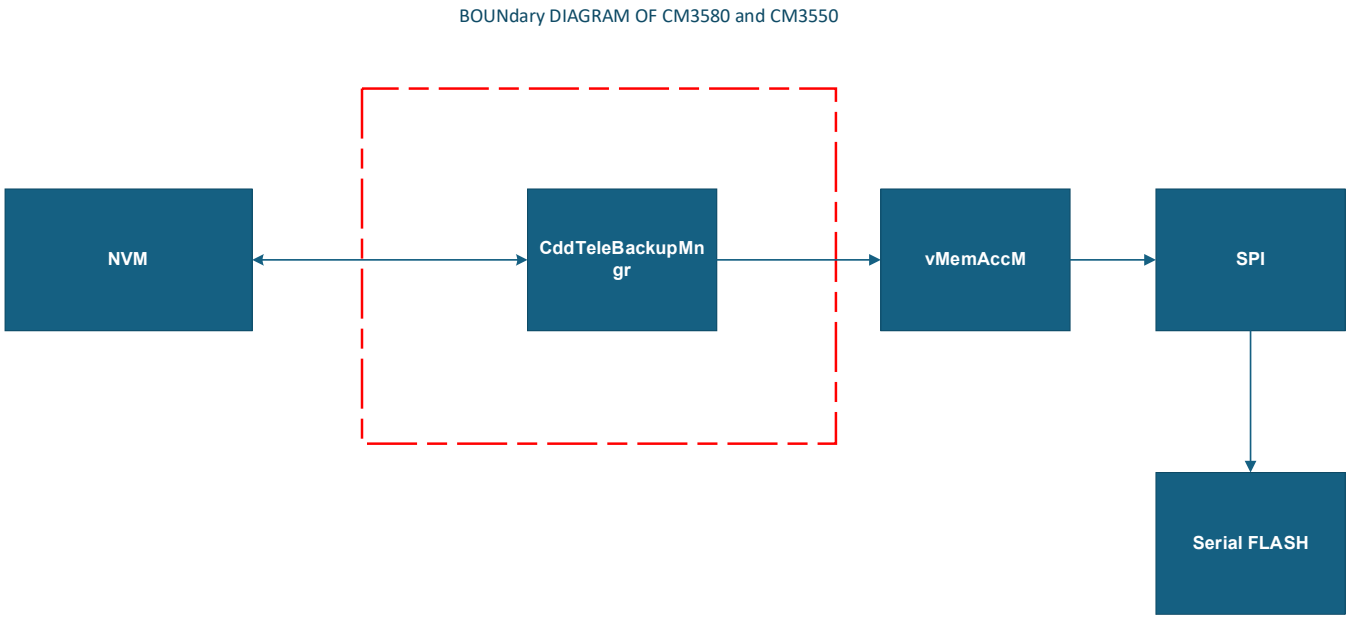
- Handle backup initiation and sequencing
- Manage secure data handling using encryption and CMAC
- Perform Serial Flash erase, write, and read-back verification
- Ensure data integrity and retry mechanism on failures
- Update persistent backup status after successful completion

The below figure shows the overall software architecture of CddTeleBackupMngr.



4.2 CddTeleBackupMngr Core Functionality or Theory of Operation

The CddTeleBackupMngr component serves as the central control module for managing secure backup operations on Serial Flash based on commands received from application-level modules via the RTE. It encapsulates the complete backup control logic and executes the operation using an internal state-machine–driven approach. Once a backup request is initiated, the component autonomously manages sequencing, data handling, and interaction with lower-level drivers and services. Program Flash data is processed in segments and blocks, with encryption and CMAC applied to ensure confidentiality and integrity. The component coordinates Serial Flash erase, write, and read-back verification to guarantee data reliability. Error conditions are handled through controlled retry and recovery mechanisms. Upon successful completion, backup status information is persistently updated in NVM and made available to the application layer.The below figure shows boundary diagram of overall CddTeleBackupMngr feature and all associated Base software components in it and the connections between them.



5. Interfaces to other components

This section shows the interfaces of CDDTeleBackupMngr with other BSW modules or software components.

Note : For Architectural view of CDDTeleBackupMngr please refer to section 4.1 General functionality in page number 9.

5.1 Provided Ports:

#	Port Type	Port Name	Port Interface Name	Data Elements	Data type	Initial Value
1	Sender	nvwBackupSfStartAddr	nviUint32	nvdVal	adtUint32	50331648
2	Sender	nvwBackupStatus	nviUint8	nvdVal	adtUint8	NA
3	Sender	nvwBufferSfStartAddr	nviUint32	nvdVal	adtUint32	33554432
4	Sender	nvwPortOverstatus	nviUint8	nvdVal	adtUint8	255
5	Sender	nvwRollBackTimerStatus	nviUint8	nvdVal	adtUint8	0
6	Sender	nvwTeleRollBackTimeOut	nviUint32	nvdVal	adtUint32	0

5.2 Required Ports:

#	Port Type	Port Name	Port Interface Name	Data Elements	Data type	Initial Value
1	Receiver	ppToolConnectedProg	piUint8	nvdVal	adtUint8	0
2	Receiver	ppTeleSpecVer	piUint32	nvdVal	adtUint32	0
3	Receiver	ppCSUFeatureEnbl	piUint8	nvdVal	adtUint8	0
4	Receiver	nvrTeleRollBackTimeOut	nviUint32	nvdVal	adtUint32	0
5	Receiver	nvrRollBackTimerStatus	nviUint32	nvdVal	adtUint32	0
6	Receiver	nvrPortOverstatus	nviUint8	nvdVal	adtUint8	255
7	Receiver	nvrBufferSfStartAddr	nviUint32	nvdVal	adtUint32	33554432
8	Receiver	nvrBackupStatus	nviUint8	nvdVal	adtUint8	255
9	Receiver	nvrBackupSfStartAddr	nviUint32	nvdVal	adtUint32	50331648

6. Requirement Tracing

Requirement	Description	Satisfied by
13584655	CSU Backup is process of coping existing calibration from program flash, encrypt it and writing it on serial flash (secured data) memory by ECM.	reCddTeleBackupSchdlr
13584660	ECM shall start its own program flash backup whenever new ECM cals/software is flashed as per "Remote Cal Download Tech Profile"	reInitCDDTeleBackupMngr , reCddTeleBackupSchdlr

13584666	ECM shall read program flash content as per following structure format provided.
	
	uint8 number_of_odfx_block;
	typedef struct
	{
	Uint32 address;
	Uint32 uncompress_size;
	
	}seg-odxf_t;
	
	typedef struct
	{
	uint8 number_of_seg;
	seg-odxf_t *seg_data[number_of_seg]; // number_of_seg gives the size of the array this points to
	}block-odxf_t[number_of_odfx_block];
		reCddTeleBackupSchdlr

13584663	ECM shall store the backup cals on serial flash	reCddTeleBackupSchdlr
13738939	ECM should calculate the CMAC on Metadata and store it to serial flash, at the end of metadata.	reCddTeleBackupSchdlr
13594046	If the backup is interrupted , the ECU shall immediately retry the operation for three times. If it fail after three attempts, set the NVM_Back_up_status to "BACKUP_FAILED". After the new key cycle, ECM shall try the backup again and set NVM_Back_up	reCddTeleBackupSchdlr
13738925	Once the backup of complete encrypted calibration of onboard flash is stored to serial flash, ECU shall set the NVM_Back_up_status to "BACKUP_COMPLT_VALID"	reCddTeleBackupSchdlr

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13738945	CMAC will be calculated for each block data, also consider each segement address (4bytes) and uncompressed length (4bytes) in Big - Endianess during generations..	reCddTeleBackupSchdlr
18134113	BSW shall disable Buffering, Backup, Port Over, Erase Memory, Rollback, Check programming dependency and respective CSU DID & RID feature when CSU_Telematics_Feature_Enable is Disabled.	reInitCDDTeleBackupMngr Rte_Read_ppCSUFeatureEnbl_deVal

7. SWC File structure

1. CddTeleBackupMngr.c
    i. Rte_ CddTeleBackupMngr.h
    ii. CddTeleBackupMngr.h
    iii. Rte_MemMap.h
    iv. CddTeleBackupMngr_MemMap.h

8. API Specification

8.1 Function List

This is a list of API functions defined in this SWC CddTeleBackupMngr.

#	Function Name	Integrity Req id	Defined in source file
1	reInitCDDTeleBackupMngr (For both CM3550&CM3580)		CddTeleBackupMngr.c
2	reCddTeleBackupSchdlr (For both CM3550&CM3580)		CddTeleBackupMngr.c
3	reTeleRollBackTimeCheck (For both CM3550&CM3580)		CddTeleBackupMngr.c
4	metadata_extract (For both CM3550&CM3580)		CddTeleBackupMngr.c

8.2 Sequence diagram for CddTeleBackupMngr

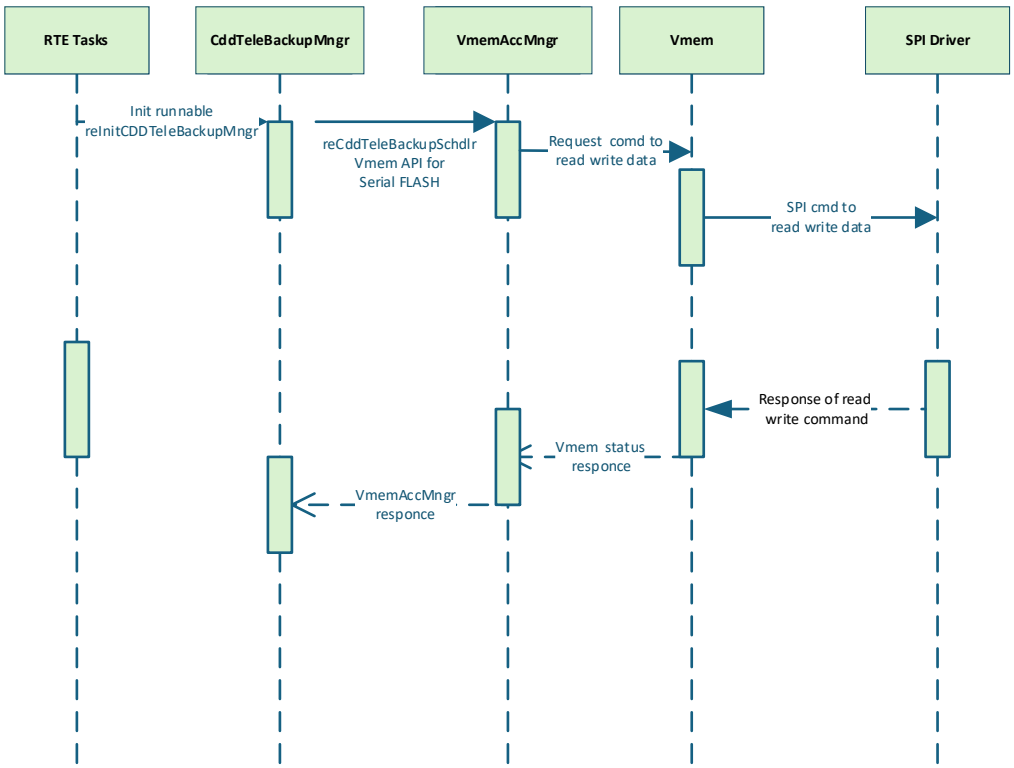


Figure 8-2-1: Sequence diagram for CddTeleBackupMngr

NOTE: To update the above sequence diagram, right click on the diagram and select ‘Visio Object’ and ‘open’.

8.3 Function: reInitCDDTeleBackupMngr

8.3.1 Function attributes:

Service name:	reInitCDDTeleBackupMngr
Syntax:	void reInitCDDTeleBackupMngr (void)
Trigger:	Executed once after the RTE is started
Parameters (in):	None
Parameters (inout):	None
Parameters (out):	None
Return value:	None

Port Access

	S-R Port Name	Data Element	Access Type	Direction
1	nvrBackupSfStartAddr	data	Explicit	Read
2	nvrBackupStatus	Data	Explicit	Read
3	nvrBufferSfStartAddr	Data	Explicit	Read
4	nvrPortOverstatus	Data	Explicit	Read
5	nvrTeleRollBackTimeOut	Data	Explicit	Read
6	ppTeleSpecVer	Data	Explicit	Read
7	nvwBackupSfStartAddr	Data	Explicit	Write
8	nvwBackupStatus	Data	Explicit	Write
9	nvwBufferSfStartAddr	Data	Explicit	Write
10	nvwTeleRollBackTimeOut	data	Explicit	Write
11	ppToolConnectedProg	data	Explicit	Read
12	nvwPortOverstatus	Data	Explicit	Write
13	nvwRollBackTimerStatus	Data	Explicit	Write
14	ppCSUFeatureEnbl_deVal	Data	Explicit	Read

Irv:	None
Description:	This function will trigger rollback timer and backup functionality

File Name	CddTeleBackupMngr.c

8.3.2 Function Design criteria and design decisions:

- Called once at ECU initialization to restore TeleBackup manager state.
- Reads NVM parameters: Rollback_timer_status, NVM_portover_status, NVM_Back_up_status.
- If NVM_portover_status indicates portover complete, it triggers rollback timer based on Rollback_timer_status.
- If NVM_Back_up_status indicates new_cal_download, it triggers backup request to start backup state machine.

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8.3.3 Function Flow chart

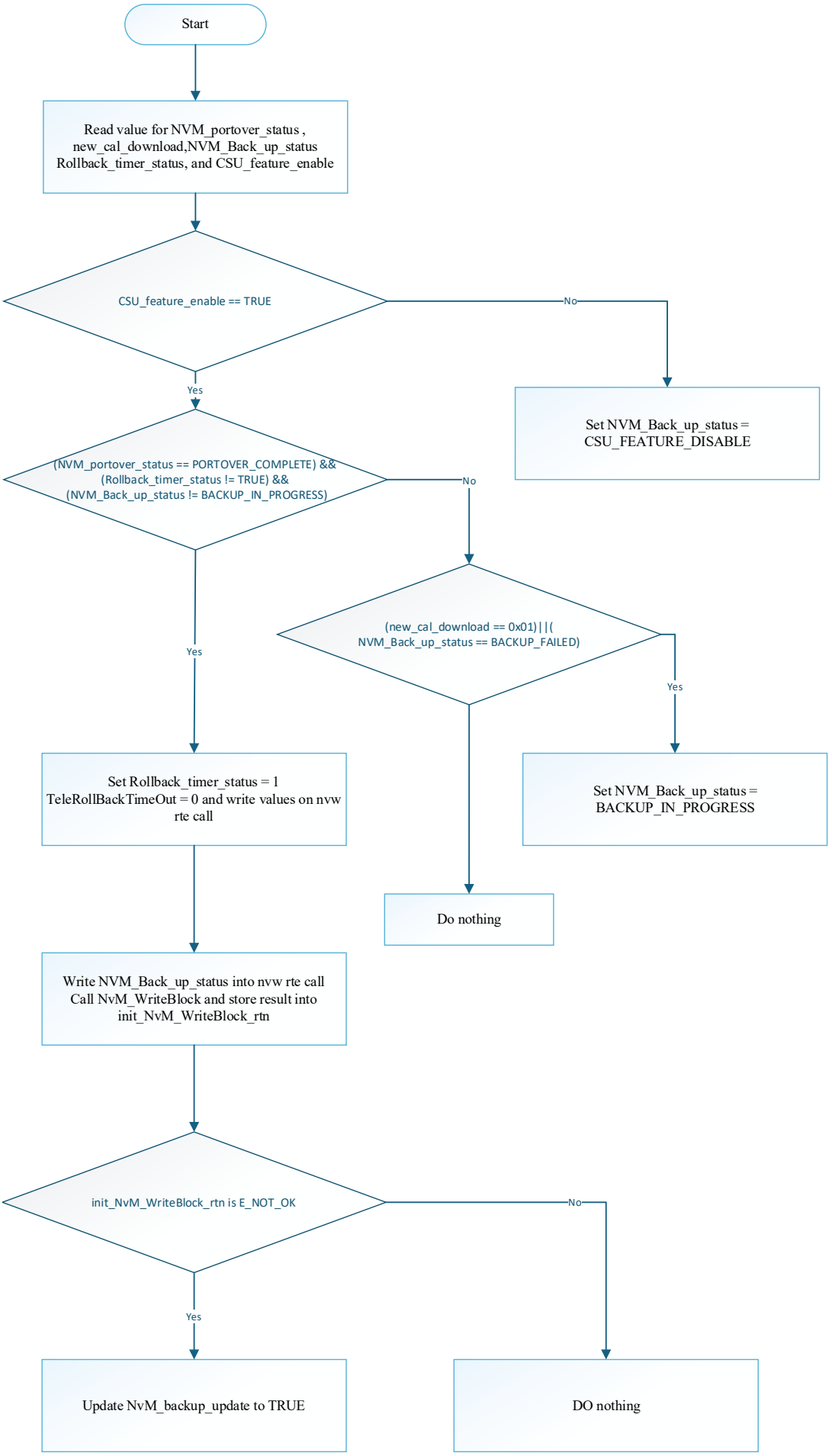

Start
Read value for NVM_portover_status , new_cal_download,NVM_Back_up_status Rollback_timer_status, and CSU_feature_enable
CSU_feature_enable == TRUE
No
Yes
Set NVM_Back_up_status = CSU_FEATURE_DISABLE
(NVM_portover_status == PORTOVER_COMPLETE) && (Rollback_timer_status != TRUE) && (NVM_Back_up_status != BACKUP_IN_PROGRESS)
No
Yes
(new_cal_download == 0x01)||(NVM_Back_up_status == BACKUP_FAILED)
Yes
Set Rollback_timer_status = 1 TeleRollBackTimeOut = 0 and write values on nvw rte call
Set NVM_Back_up_status = BACKUP_IN_PROGRESS
Do nothing
Write NVM_Back_up_status into nvw rte call Call NvM_WriteBlock and store result into init_NvM_WriteBlock_rtn
init_NvM_WriteBlock_rtn is E_NOT_OK
No
Yes
Update NvM_backup_update to TRUE
DO nothing

Figure 8-4-1: Flow chart for Function reInitCDDTeleBackupMngr ()

NOTE: To update the above sequence diagram, right click on the diagram and select 'Visio Object' and 'open'.

8.3.4 Function Call Tree

Note : For Sequence Diagram please refer to section 8.2 and 8.3

8.4 Function: reCddTeleBackupSchdlr

8.4.1 Function attributes:

Service name:	**reCddTeleBackupSchdlr**
Syntax:	void **reCddTeleBackupSchdlr**(void)
Trigger:	triggered on TimingEvent every 10ms
Parameters (in):	None
Parameters (inout):	None
Parameters (out):	None
Return value:	None

Port Access:

	S-R Port Name	*Data Element*	*Access Type*	*Direction*
1	nvrBackupSfStartAddr	data	Explicit	Read
2	nvrBackupStatus	Data	Explicit	Read
3	nvrBufferSfStartAddr	Data	Explicit	Read
4	nvrPortOverstatus	Data	Explicit	Read
5	nvrTeleRollBackTimeOut	Data	Explicit	Read
6	ppTeleSpecVer	Data	Explicit	Read
7	nvwBackupSfStartAddr	Data	Explicit	Write
8	nvwBackupStatus	Data	Explicit	Write
9	nvwBufferSfStartAddr	Data	Explicit	Write

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	10	nvwTeleRollBackTimeOut	data	Explicit	Write
	11	ppLogCSEvent_LogEvent	uint8 EventType, const uint8 *ReasonCode, uint16 dataRec1, uint16 dataRec2	Explicit	Client
Irv:	None				
Description:	This function is used to execute state machine for backup functionality.				
File Name	CddTeleBackupMngr.c				

8.4.2 Function Design criteria and design decisions:

This runnable is invoked every **10 ms** and acts as the main scheduler for the telematics backup manager when a backup has been triggered.
It runs only if the backup status equals BACKUP_IN_PROGRESS and the CSU feature is enabled – otherwise it returns immediately.

Core responsibilities:

1. **State-machine control:**
   - Erase the serial-flash region, determine the next PF address, and manage per-segment/page processing.
   - Drive the crypto sequence (IV generation, encryption start/update/finish and CMAC start/update/finish).
   - Write encrypted pages to flash, read back for verification, and handle block-level CMAC storage.
   - Manage retry logic (up to three attempts) and transition to BACKUP_COMPLETE or BACKUP_FAILED.
2. **Metadata handling:**
   - Extract metadata from telematics_csu_table, build an array, compute its CMAC, and write it to flash.
3. **NVM updates & logging:**
   - Write NVM variables for status and timers.
   - Trigger NvM_WriteBlock when status changes.
   - On permanent failure, log a cybersecurity event using Rte_Call_ppLogCSEvent_LogEvent.

8.4.3 Function Flowchart

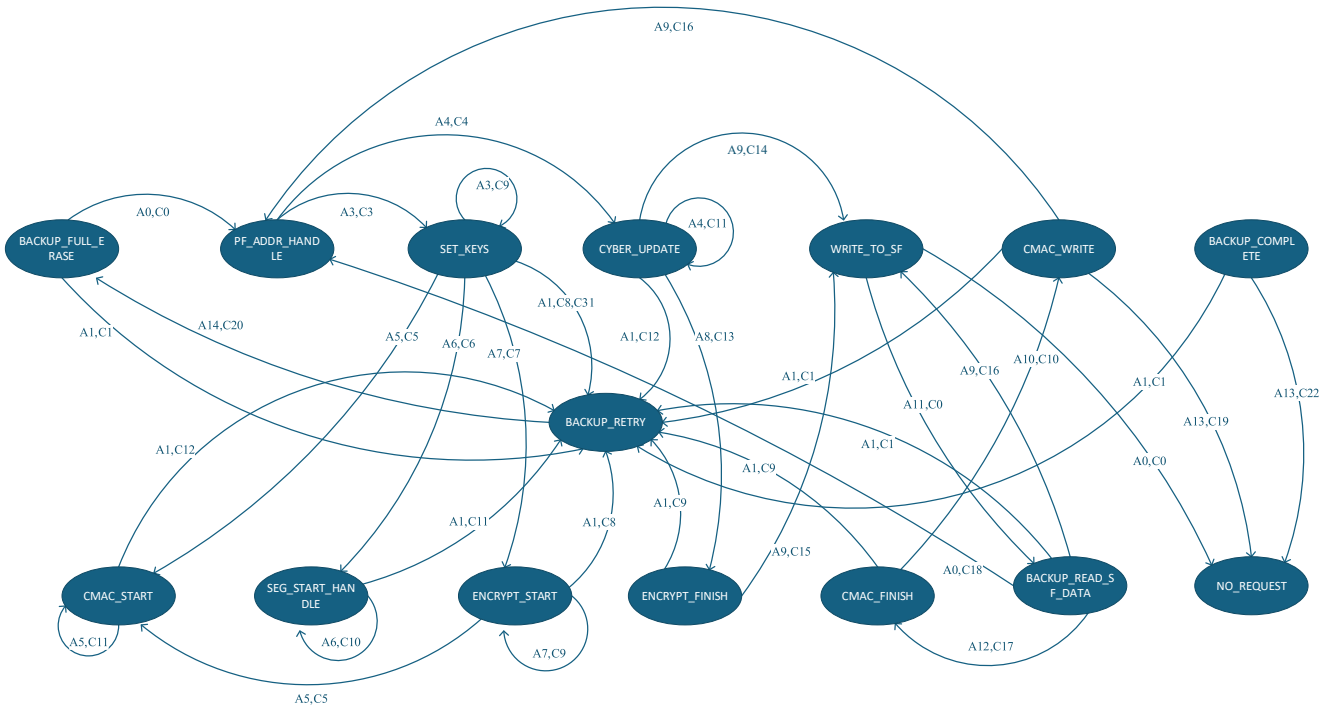


Figure 8.4.3-1: Flow chart for Function reCddTeleBackupSchdlr **()**

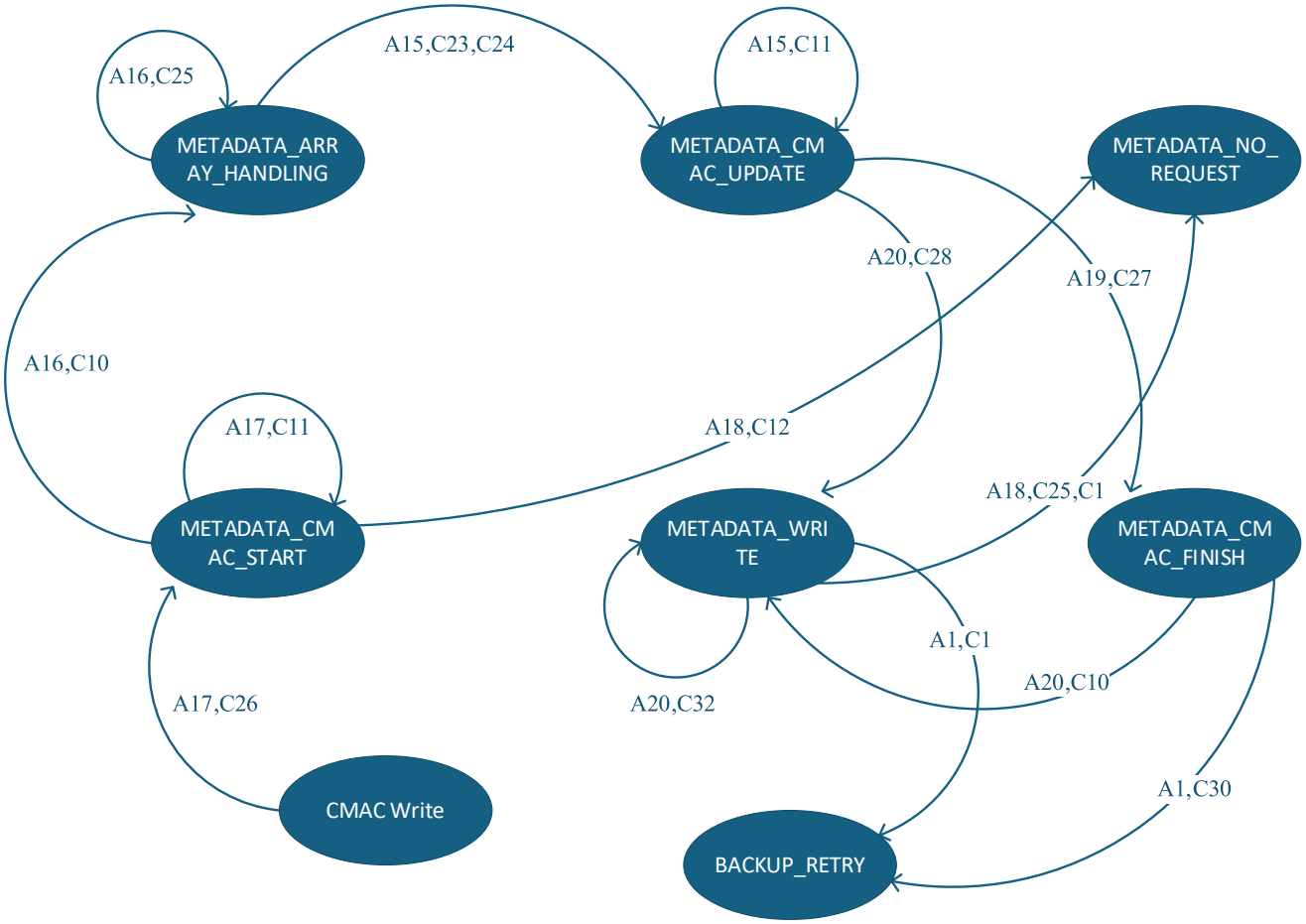


Figure 8.4.3-2: Flow chart for Function reCddTeleBackupSchdlr **()**

NOTE: To update the above sequence diagram, right click on the diagram and select ‘Visio Object’ and ‘open’.

Actions	Description
A0	Go to PF_addr_handle state
A1	Go to Backup retry state
A2	Go to encrypt_start state
A3	Go to set_keys state
A4	Go to cyber_update state
A5	Go to CMAC_start state
A6	Go to seg_start_handle state
A7	Go to encrypt_start state
A8	Go to encrypt_Finish state
A9	Go to write_to_SF state
A10	Go to CMAC_write state

A11 Go to backup_read_sf_data
A12 Go to CMAC_finish
A13 Go to no_request state
A14 Go to full_erase state
A15 Go to metadata_cmac_update
A16 Go to metadata_array_handling
A17 Go to metadata_cmac_start
A18 Go to metadata_no_request
A19 Go to metadata_cmac_finish
A20 Go to metadata_write state

Conditions
C0 return of serial flash state is E_OK
C1 return of serial flash state is FAIL
C2 return of serial flash state is E_NOT_OK
C3 return status Csm_RandomGenerate is E_OK
C4 PF reached to read last page of segmengt
C5 If Segment Index is 0
C6 return of set_keys is E_OK
C7 return of encrypt_start is CRYPTO_E_BUSY
C8 if Csm_encrypt return ≠ E_OK and not CRYPTO_E_BUSY
C9 return status of Csm_encrypt is not E_OK
C10 return status of Csm_macGenerate is E_OK
C11 return status of Csm_macGenerate is crypto-e-busy
C12 return status of Csm_macGenerate is not E_OK and not CRYPTO_E_BUSY
C13 last page of segment and backup-result len <= page-size
C14 when it’s not last segment last page
C15 CMAC_ENCRYPT result is E_OK
C16 padding length = 10
C17 PF_addr == block end addr
C18 PF_addr ≠ block end
C19 C0 and last block_CMAC ==1
C20 if backup attempt ≤ 3
C21 if backup attempt ≥ 3
C22 serial-flash state is SF_REQUEST_COMPLETE or SF_NO_REQUEST
C23 for every new CSU header
C24 metadata segment break == 1 and metadata block break == 0

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C25 (backup_data) > &backup_matadata_array[SF_PAGE_SIZE - 1]

C26 serial-flash state not fail and last-block_CMAC == 1

C27 C10 & metadata_complete == 1

C28 C10 & metadata_complete != 1

C29 C0 & metadata_complete != 1

C30 return of Csm_mcaGenerate is not E_OK

C31 return status of Csm_keySetValid is not E_OK

C32 backup-data > backup-metadata-array[SF_PAGE_SIZE-1]

8.4.4 Function call tree:

Note : For Sequence Diagram please refer to section 8.2 and 8.3

8.5 Function: reTeleRollBackTimeCheck

8.5.1 Function attributes:

<table>
<tr><td>Service name:</td><td>reTeleRollBackTimeCheck</td></tr>
<tr><td>Syntax:</td><td>FUNC(void, CDDTeleBackupMngr_CODE)
reTeleRollBackTimeCheck(void)</td></tr>
<tr><td>Trigger:</td><td>triggered on TimingEvent every 5s</td></tr>
<tr><td>Parameters (in):</td><td>None</td></tr>
<tr><td>Parameters (inout):</td><td>None</td></tr>
<tr><td>Parameters (out):</td><td>None</td></tr>
<tr><td>Return value:</td><td>None</td></tr>
<tr><td>Port Access</td><td>
<table>
<tr><th></th><th>S-R Port Name</th><th>Data Element</th><th>Access Type</th><th>Direction</th></tr>
<tr><td>1</td><td>nvrRollBackTimerStatus_nvdVal</td><td>data</td><td>Explicit</td><td>Read</td></tr>
<tr><td>2</td><td>nvrTeleRollBackTimeOut</td><td>data</td><td>Explicit</td><td>Read</td></tr>
<tr><td>3</td><td>nvwRollBackTimerStatus</td><td>data</td><td>Explicit</td><td>Write</td></tr>
<tr><td>4</td><td>nvwTeleRollBackTimeOut</td><td>data</td><td>Explicit</td><td>Write</td></tr>
</table>
</td></tr>
<tr><td>Irv:</td><td>None</td></tr>
<tr><td>Description:</td><td>Runs every 5 s.
If the rollback timer is active it:
• Increments the stored timeout,
• Compares against the calibration caTeleRollBackTimer,
• When expired,
sets NVM_Back_up_status to BACKUP_IN_PROGRESS and clears the timer flag.
Writes updated NVRAM values and requests an NvM write block.
No action taken when the timer is inactive</td></tr>
<tr><td>File Name</td><td>CDDTeleBackupMngr.c</td></tr>
</table>

8.5.2 Function Design criteria and design decisions:

This runnable is executed every **5 seconds** via a timing event.
Its job is to manage the rollback timer that controls when a telematics backup should automatically begin.
If the rollback timer is active it:

- Increments the stored timeout,
- Compares against the calibration caTeleRollBackTimer,
- When expired, sets NVM_Back_up_status to BACKUP_IN_PROGRESS and clears the timer flag.

Writes updated NVRAM values and requests an NvM write block.
No action taken when the timer is inactive

8.5.3 Function Flow chart

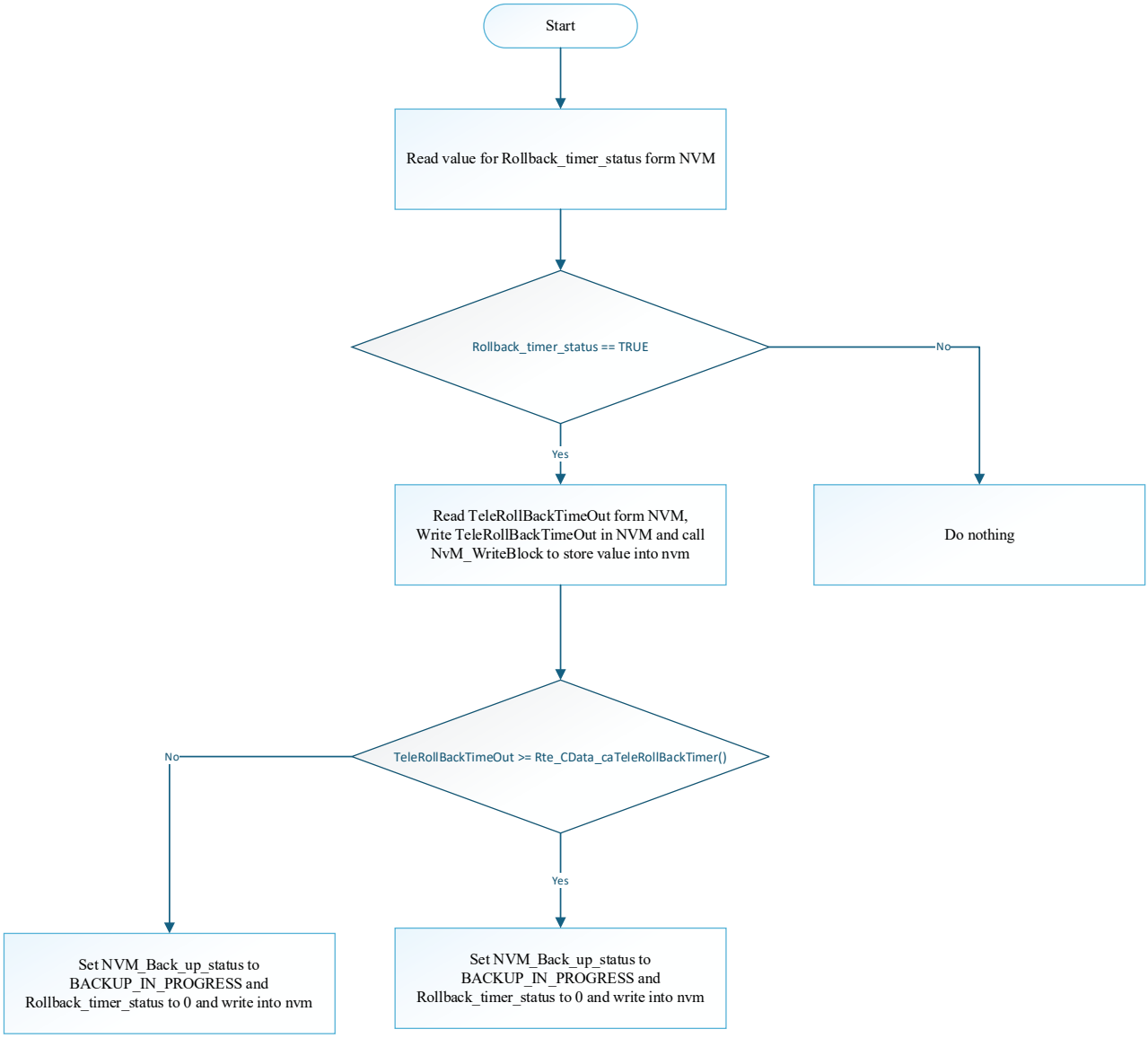


Figure 8-6-1: Flow chart for Function reTeleRollBackTimeCheck()

NOTE: To update the above sequence diagram, right click on the diagram and select ‘Visio Object’ and ‘open’.

8.5.4 Function Call Tree

Note : For Sequence Diagram please refer to section 8.2 and 8.3

8.6 Function: metadata_extract

8.6.1 Function attributes:

Service name:	**metadata_extract**
Syntax:	void **metadata_extract**(void)
Trigger:	Call form reCddTeleBackupSchdlr
Parameters (in):	None
Parameters (inout):	None
Parameters (out):	None
Return value:	None
Port Access:	None
Irv:	None
Description:	This function initializes the IoHwAb_HBDiagBuffer (including the rety limit).
File Name	tc3xx_hbridge.c

8.6.2 Function Design criteria and design decisions:

Extract CSU Metadata Structure – Populates the csu_header_backup structure by copying block count, segment count, source program flash addresses, and uncompressed data sizes from the telematics_csu_table.
Calculate Metadata Length – Computes total metadata size including fixed header length, block headers, segment data structures, and CMAC (16 bytes), accounting for all blocks and segments.
Determine CMAC Reference Address – Calculates CMAC padding and derives the serial flash reference address (block_cmac_addr_reference) for storing block-level CMAC values during backup operations.

8.6.3 Function Flowchart

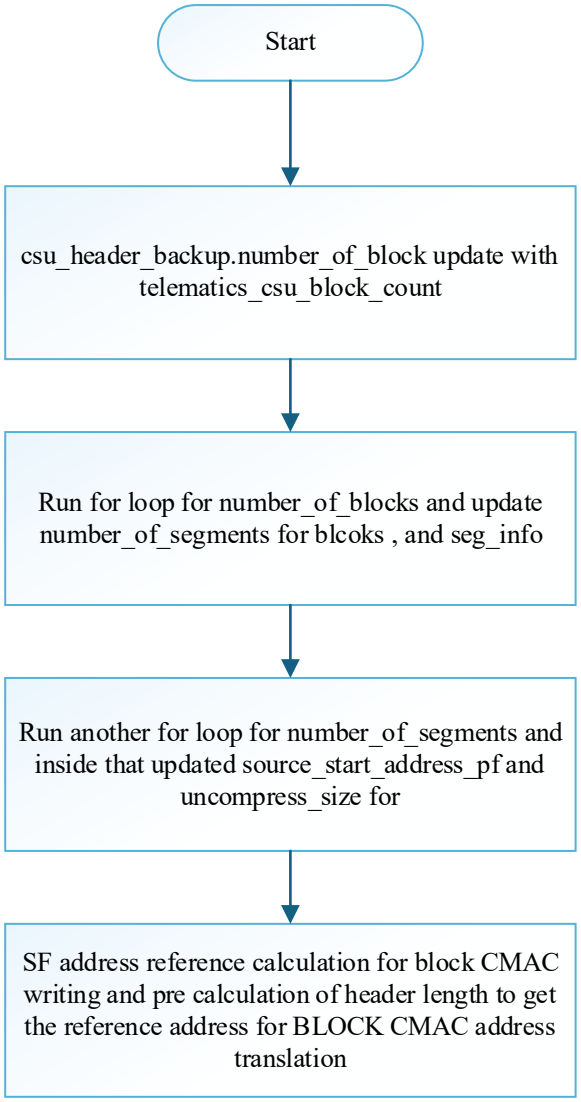


Figure 8-8-1: Flow chart for Function metadata_extract ()

NOTE: To update the above sequence diagram, right click on the diagram and select ‘Visio Object’ and ‘open’.

8.6.4 Function call tree:

Note : For Sequence Diagram please refer to section 8.2 and 8.3

8.7 Error classification

8.7.1 Development Errors

None

8.7.2 Production Errors

There are no production errors. Lists DEM Events to be supported and they will be mapped to DT

9. Datatype Mapping

Following are details of implementation Datatypes associated with Application Datatypes used in Port's data elements:

	Application Datatype	Implementation Datatype
1	adtBool	boolean
2	art piLogCSEvent	irtpiLogCSEvent
3	adtsint32	sint32
4	adtUint32	uint32
5	adtUint8	uint8

10. Constants

In general, this chapter captures the calibration parameters, software constants, Nv data and Macros used in the SWC CDDTeleBackupMngr.

10.1 Calibration Parameter:

Following are the calibration parameters used in implementation.

Calibration Name
Rte_CSUFeatureEnbl

10.2 Software Constants:

Following are the Software constants used in implementation.

- None.

10.3 NV DATA:

Following are the Software constants used in implementation.

- None.

10.4 Macros:

Following are the macros used in implementation.

Macro	Default Value
Cipher_IV_LENGTH	16
CMAC_GEN_LENGTH	0x10
BLOCK_CYBER_HEADER_LENGTH	0x188
CSU_METADATA_STRUCT_FIX_LENGTH	0x17
SEGMENT_DATA_STRUCT_LENGTH	0xC
DBG_TELE_CSU_BACKUP	STD_ON
NULL_TELEBACKUPMNGR	((void *)0)
BACKUP_DATA_SIZE	10U
SEG_INFO_SIZE	8U
CDDTBM_SEG_INFO_SIZE	8u

11 Lessons Learnt

This section is to capture any lessons learnt during previous design cycle of this SWC. The intent is to not make the same mistakes in future.

12 Revision History

Revision history of this document is shown in below table.

Revision Level	Date	Name	Description
1.0	09-03-2026	Pandurang Misal	Initial creation of document.